

ActInf Textbook Group ~ Cohort 1 ~ Meeting 4

(Chapter 2 pt. 1)

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hello everyone it's may 26 2022. it's the fourth week of the first cohort of the octave textbook group we're in our first discussion of chapter two today we're going to go over the questions that are posted for chapter two then we'll see where that takes us and what ideas of the book have been explored and that'll be the main focus also just a note on the math learning group those meetings have been occurring on 19 utc on wednesdays and so in the math learning group you'll find information about how to participate you'll find and be able to edit resources on learning different topics so this is a really helpful activity if people want to share the resources that help them learn about different ideas and that will help people search for a video about this we have a notation table that will help us as we start to look at the notation more there's some math learning basic questions like around math anything related to the textbook content can go in the main questions math learning group is part of the real textbook group so these are not the questions to be answered like the other ones that we're going to go to are but this is just about math and then also there's some math oriented overviews like i wrote this summary of the first two-thirds of chapter two so if people want to be like continuing to distill and represent what they're learning and what they're asking that's all good okay any other um notes that anybody wants to add before we start to go into the questions you can raise your hand or you can type it in the chat so we're in the first week of discussing chapter two the low road to active inference so just to kind of warm us up what is the low road [Music] yeah the low road is uh the bayesian brain approach right here um is the like proximate next like big concept i guess previous but it's kind of based on i guess yeah i mean you could see right in the picture like it's based on basic it's like mathematical i the way i conceptualize it is like literally you know the low road meaning like um very primitive very mathematical very like hyper analytical foundational kind of you know an analysis way of looking at it i guess that's not necessarily living or anything like that right whereas the high road is kind of more about the other side of that right so we're going to be learning about the high road and low road those

are the two paths that are offered in chapter two and three of the book so hopefully again like it would be awesome to have as many people as possible speak and feel welcome to share their views during this session so everyone is especially encouraged to put questions or comments in the chat or raise their hand to speak or find other times to have synchronous communications and especially to make contributions to the coda but really appreciate that everyone has come out to go through these discussion questions so we're going to now go to the that have been placed into the questions table and just the more questions the better they can be super short it can be anything that you're having uncertainty if you come across a table or a figure or sentence in the text and you have a question please write the question down it will be extremely extremely helpful it doesn't have to be with extensive references or citations it can just be what did this mean if you're uncertain about something and then if you were certain about it you could still write a question like what would you ask somebody to test or to understand what they understood about it okay so we're gonna start with this first most upvoted question but there's still time in this session and also till next week in our next session on chapter two to add more questions if you wrote them down somewhere else and also to upvote questions but we're just going to start with the most interestingly voted questions what is the relevance of the support and surprise for the different probability distributions in table 2.1 so what does anyone see in table 2.1 and what are support and surprise people mic yes so table 2.1 looks to me excuse me like a guide for selecting distributions um if you're building out one of these models with support saying uh you know what set of numbers would fit within this distribution selection and then if you chose this distribution how would you look at the support equation for it yeah so if you chose this distribution how would you look at the surprise component of it and we're going to get to surprise and busy in surprise in the coming questions so support using the definition pulled here from wikipedia the support of the function is the subset of the x-axis like the domain where there's a y value that's not zero so it's where the function is non-zero defined the gaussian the normal bell curve distribution has non-zero support at all numbers fancy r and then other distributions have different variable different values that they're having non-zero values at like the gamma distribution is defined over the interval from zero to infinity so if it was like how many of this thing do you have that might be a reason to use a that's only positive numbers or if you were wanting something that could have positive or negative numbers of any size the gaussian is a and then another one that's like a relevant support difference would be like the dear schley being defined only within the zero to one interval which helps connect it to probabilities so the support is where

that function that distribution family is defined and then what is the surprise does anyone know the name of the little squiggle they are using to represent surprise i think it's called fracture i aka fancy eye the first we're going to come to the definition of surprise in a coming but um and without going into every variable and trying to overlearn something the surprise of a given observation like a data point let's just think about the gaussian x is the observed data point and μ is like the mean of that normal distribution which is also the mode so it's like the center of that and then this capital π is a variance estimator i believe here so how surprised one should be by a new data point coming in is related to the difference between that new data point and the mean so if the new data point were exactly on the center of the distribution this whole term is going to go to zero so one should not be surprised at all by exact data points on the mean of the distribution but any data point that has not exactly on the mean is going to have some non-zero value and so it's going to be a function of how far the data point is from the mean and some scaling by the variance so if we were expecting 100 with a variance of 10 versus 100 with a variance of one in one case like getting 101 is like a one sigma event it has a z score of one just speaking loosely here whereas in the um case of it having a variance that's very small then that might be more surprising comments that people have on table 2.1 uh i have a question set while i was reading through the chapter uh there so okay uh going back to section 2.2 right perception as inference so when he says when they say that the brain is a protective machine or a statistical organ that in first affects external states of the world okay here's my question because i'm primarily from a telecom background most of our things are to get messages across between two distance transmitters uh receiver right so we build noise models so without any noise you don't get signal so you need to know what a noise model looks like and then you know so for example if the power spectrum of the signal is what you expect from let's say you're using a distribution of noise right so if it's the same as you expect from the gaussian uh you do nothing so that's just noise so you don't classify the data that's coming in if it is more than this then uh okay so there might be something interesting about this there might be some signal here so we define the signal to noise ratio a lot of this chapter seems to be dealing with the fact that we by default know what the signal looks like so there's no attempt to build a noise model here that would be like okay so uh maybe the sensor in this case our eyes for example or has some parameters for it right it can detect between say a bandwidth well okay so it can detect a light between certain frequency ranges but maybe it performs very well in a certain frequency range and not others there's more error in other frequency ranges right so the noise model should incorporate such imperfections in the sensor

imperfections in the connections between the sensor and processing equipment but but there's no attempt to do that in this entire chapter so i'm just curious as to what exactly is the predictive machine here what what is being predicted here great question and in general every question that people have they're going to be 100 times more impactful if they can be written down this is just the second chapter it's introducing us to the essence of starting with just bayes equation it's the first equation here starting with the the foundation of increasingly advanced noise filtering approaches so check out live stream 43 to see how calm and filter and advanced multi-level bayesian filtering schemes are used and learned and fit with free energy minimization and so on but you're right that is not approached in this chapter because it's starting with the essence of bayes equation and then going to build towards another kernel which is going to be the kernel of the variational free energy and then that in this one layer case is going to be able to be elaborated and parameterized and fit within nested models to accommodate like learning on hyper parameters learning of noise models and so on but it is not in this chapter okay yeah so i was just i'm sorry if i threw us off track no so it's perfect but they're good questions and then jakob asked what do you think of the surprise the distributions involving a sum product of the product probability distributions except for gamma actually i guess the gaussian kind of as well yeah maybe i'm thinking about this wrong but uh i thought that the surprise was like the i guess the information gain um or equivalent and so for the for the gaussian and the gamma distributions it's always for one single one single element uh of the distribution for one x but for the multinomial and dirichlet there's something over all of the possible probabilities which makes it i guess an entropy function is there is there a difference for example one could have a multi-dimensional gaussian a two-dimensional gaussian and then the surprise for the would be represented as a sum over the different dimensions so they've shown the gaussian and the gamma in the uni-dimensional case and the multinomial in the dirichlet in a multi-dimensional case so in that case is it i thought that x was just one element of the domain so that's not necessary so x is just if we think of the gaussian as this kind of bell shaped curve on the x and y axis we really want the f of x value to get like the gaussian value but x is just the input value is that correct leave that as accurate because yes x is the domain of the function and then the sum is over eyes so if we had a multinomial gaussian or multi-dimensional gaussian in eye dimensions then it would have a sum over eyes yeah so if but wouldn't uh i guess i still have uncertainty about why we don't do that for the single uh single-dimensional gaussian as well because if we want to if we take just one point on the gaussian i understand why that would have a certain surprise but then in the same way

if we take one point in the dirichlet wouldn't that also shouldn't that also have an equivalent equation for surprise why why in the dirichlet dirichlet distribution are we summing over all the i's but not for the but not but in the gaussian we are considering just one point on the probability distribution or maybe is maybe dirichlet is multi multi-dimensional and that's the reason i think this the uni-dimensional case would be just without this sigma i it would just be this but if anyone has any other thoughts especially if they want to contribute them in like writing to improve our understanding that'd be awesome let's continue to next question in section 2.4 they write we now introduce the simple but fundamental advance offered by active inference this starts from the same inferential perspective discussed above but extends it to consider action as inference in your own words what is the simple but fundamental advance offered by octave so what would anyone write or like to share what is the advance of active inference so is it uh just bringing action into the equation predictive coding and bayesian brain are specifically about perception visual visual perception um i think the key thing with active inference is that it's um well it's not just that it brings action into the equation it's that it treats them as essentially the same thing that that kind of united under as part of the same process which i think is which i think is kind of a new development thanks for the awesome answer ali uh yeah i think uh alan barthold or bartholos in the brain's sense of movement uh is a has a particularly uh relevant um description of action and perception and uh he writes that perception is simulated action and i think that really speaks to uh the advance made by uh active inference i mean uh the integration of action and perception uh or uh in other words uh as barthold claims the simulation of action and perception i guess a really novel development i think i read something but i i can't remember where i read it but i i think it was roughly that on these existing bayesian brain theories an aid an agent perceives in order to act whereas under active inference and agent acts action kind of precedes perception so it active inference essentially flips this traditional schema in terms of which one feeds into the other yeah i was just gonna say that uh yeah adding on to bringing action into the equation it flips it like uh there's no where to where do your priors come from where do the hyper parameters get specified like where does the model start where does where is the agency kind of in all of these other models it's really uh no accordance for that or it's hard to it's like very emergent because here it's like very explicitly specified some affordance for that thanks brock uh mike and then anyone else who raises their hand yeah so i'm thinking if you have priors you have some prior assumptions then do you necessarily need to take action in order to sort of envision what your perception might be so it's almost like getting into imagination or something like

that you have a set of priors that can drive uh what you imagine your perception to be before any action is taken great question blue so i think um that this was like elaborated on a lot in section 2.7 when they talk about um pursuing a policy um because there's consequences like when we're planning and making predictions that has consequence consequences for both our perception and our action um like when i mean even just planning even in a conversation when someone says oh have a nice like have a nice day as expected at the end of that but like have a nice rabbit would be like what like you're you're not expecting that that to come at the end of that that sentence so so when you plan it interferes with your perception because you didn't plan for them to say have a nice rabbit right but it also interferes with with your um action and so i think like this the um the like novelty or enhanced aspect of active inference loops into the fact that perception and action are continually impacting one another yes thanks blue these are some of the core memes and themes and it really is important to understand how active inference is similar and different than other work in this area like variational bayesian inference is not introduced by bayesian models of perception is not introduced by active inference bayesian models of action is not so it's about finding what has been done to understand what is being offered and then whether one chooses to take this like history of science development of science view or we kind of just want to state plainly what it is without wondering what the advance relative to other frameworks is these are all really critical ideas like the signal processing framework was brought up earlier and predictive processing predictive coding anticipatory systems are often purely about sense just to give one thought on that it kind of makes sense for a video encoding algorithm for example because every piece of the camera is in focus vision requires action and so that's where we're going to be bringing in all these other important concepts attention sensory attenuation the active decision making component of vision to resolve uncertainty which is what gives rise to a generative visual field that seems like there's color everywhere and seems like there's high resolution everywhere though that is not the incoming sensory information by fully taking this generative modeling perspective on perception which um in live stream number 43.0 maria did an awesome job of connecting this even before helmholtz and kant it's plato's cave and it's part of this long discussion about perception action could be a variable there's perceiving and then if action is a variable then active inference is providing not just a unified framework like a bayesian graph framework for some variables that are interpreted as perception and some variables that can be interpreted as action selection or policy planning there's a tractable approximation by shifting from the in bringing this answer by shifting from an exact solution to the mathematical

problem bayesian inference to an approximate solution variational free energy which is going to provide a bound on some quantity that might be intractable to compute so it's going to give a heuristic for the action perception cycle but this is a really great question so people can continue to return to it because people will probably ask us for a long time to come how is this different than blank or what is active inference these are some of the things that we would want to have in mind ali yeah again going back to bartow his definition of perception as i think it uh also relates nicely to uh marla ponti's phenomenology marlopanti uh has a famous statement and uh as uh has a famous insight as vision is the brain's way of touching uh and so you see uh for instance we as we move around space as we uh construct our sense of spatiality uh i mean or our sense of temporality or everything else uh we don't just begin with the representation of space or time uh i mean the image of movement uh constructs because we move and not as a consequence of our movement thanks and we'll continue with the questions but that's an awesome area to go into with phenomenology with 4e cognition extended embedded in culture et cetera et cetera and many many of the live streams and papers have been on those areas so it's cool to bridge from formal models of perception cognition and action into qualitative and philosophical areas and then oh thanks for adding that the idea of interrelated action and perception has been around in cognitive science for a long time dynamical systems theory and inactivism emphasize that but active inference brings a coherent formalism to the table which is a nice advance well said next question figure 2.2 figure 2.2 illustrates why as a result of an internal generative model x and an external generative process x^* in order to measure surprise wouldn't we need another value of y i.e a separate y that encodes prior beliefs if y can be objectively measured from external signals is there a third y that's considered the observation first i'd like to actually go to this question about two notions of surprise so um because they might come to bear on this action perception loop once we move from just bayes equation and calculating surprise on parametric distributions to thinking about prior updating and cognitive entities with the generative model that constitutes a prior with incoming information coming in we're going to start to see why this notion of surprise and bayesian surprise are similar and different so using the apple frog example so the unobserved hidden state the latent state of the world is whether there is an apple or a frog in a bag for example or just in this person's area what can be observed just speaking coarsely is it can jump or not and we can totally go down the rabbit hole with what is truly observed and things like that but this is just what are in the context of this model the unobserved state is the actual identity of the object and the observation is going to be the action or not we take the

opportunity to unpack two different notions of surprise both of which are important the first we refer to simply as surprise it's the negative log evidence where evidence is the marginal probability of observations so we saw that first sense of surprise with the fracture i the previous table we looked at the second notion of surprise is referred to as bayesian surprise this is a measure of how much we have to update our beliefs following an observation in other words bayesian surprise quantifies the difference between a prior and a posterior probability what are the similarities and differences between the two notions of okay we'll see what has been said and then hear what other people are thinking um so page 20 they wrote that bayesian surprise scores the amount of belief updating as opposed to surprise which is simply how unlikely or likely that observation was so in that gaussian case the surprise is like about one data point coming in given how the distribution's parameterized right now how surprising was that one data point and then bayesian surprise is going to be about how much that distribution is updated after processing that data point similarities they both depend on how well the agent's generative model matches the external world and they're both measuring surprise or bayesian surprise in the same units which are information theoretic quantities of information i.e gnats or bits differences plain surprise marginalizes over all the models degrees of freedom under the model's prior distribution over its adjustable parameters that would be great for someone to unpack what does it mean to marginalize over the model's degrees of freedoms and then bayesian surprise lets the model choose the best set of parameters it can to fit the observed data measures how much of an update that was um mike and then and then blue and then only yeah under the first similarity i'm wondering if um it's this dependency on how well the agent's generative model matches their perception of the external world as opposed to matches the external world agree with that great addition blue and then only okay i think my hand was just left over left up okay ellie uh well i believe that the plain surprise is a kind of raw statistical surprise but bayesian surprise in a as a kind of processed surprise or i'm not sure if i'm right in saying that plain surprise can probably be described as an objective surprise as opposed to bayesian surprised as a subjective surprise but as i said i'm not sure about the plausibility of these descriptions great comments um even surprise alone still depends on the parameters of the generative model so it's still is within a processing or a filtering frame albeit a fixed one so um they talk about like uh let's let's look at the figure where there's a graphical overview of this apple and the frog and this is kind of giving some graphics and words to fill in this apple frog example so initially the person has this likelihood model in the back of their head where they have beliefs about how likely apples are to jump they do it one percent of the

time and how likely frogs are to jump they do it 81 of the time that's the likelihood that's about how observations depend on hidden states of the world their prior beliefs are that there's a 10 percent chance that the entity is a frog and these are mutually exclusive options there's no third option here that would be like another category um model structure learning on the model so we're staying within this model for now and they sum to one because they're a probability then there's an observation which is jumping the posterior reflects the updated beliefs about what the entity is and so it's like it's so much overwhelmingly more likely that frogs jump that seeing something jump updates the prior from here to here so in this case one can calculate the surprise of the observation without doing any updating at all one could just stay fixed in their prior belief could describe how surprised they are in gnats by given observation in this full bayesian cycle there is an updating of the prior to the posterior and then that updating can be described in terms of how much the prior was updated and so that's like here the model to the observed data this is now the best fitting model and then we're calculating the difference between these two um regular surprise being zero means the data point was exactly as you expected bayesian surprise being zero means the distribution was not updated high surprise means that the data point was extremely unpredicted it was extremely unlikely whether or not you update your model at all high bayesian surprise means that the distribution was changed a lot as a function of seeing that happen uh well perhaps i didn't understand it correctly but uh isn't it that isn't the case that in uh the plane surprise or uh more generally in the uh the bayesian formulation of the probability of events uh the likelihood and both the likelihood and the priors are uh somehow inherent to the phenomena inherent to the events independent of the observers a great question i hope i'm not going off on a branch here but this is related to the difference between frequentist and bayesian approaches to statistics so frequentism does have viewed from the bayesian perspective frequentism does have priors they're uniform priors which are not uninformative priors they just are uniform and so in frequentism we come across like the maximum likelihood solution or the maximum likelihood parameter which is just like well if all outcomes were equally a priori likely uniform prior we would just need to evaluate the and find the model with the maximum likelihood solution bayesian offers another degree of freedom and says well sure you could pick a uniform prior that would give you the maximum you might want to have a prior over that space so that if something is twice as likely a priori and then you observe less than 2x evidence for it you still might want like the bayes factor or your posterior to reflect that one rather than just jumping instantly to a different maximum likelihood solution so yes priors and likelihood are implicit but they're using a

different ontology than bayesian statistics we're in the bays or the post-based area now but there's so many connections to classical statistics and in spm the textbook there is parametric classical statistics non-parametric classical statistics and bayesian statistics they're more similar than not it just is about seeing where one of them is like a special case of another or a generalization of another let's see if we can do one or two more questions during this session okay so let's return to the previous so we're looking at figure 2 2 which is something we're going to see different representations of entity as generative model world or niche as generative process and then here the hidden states are those that are unobserved as data y is referring to data points that are observed as data there's a cognitive hidden state which is like a prior in this generative model and then there's some hidden state but it could have been any letter or any shape about the hidden state in the world the observation is going to everything that happens in between here is cognition um like the sandwich model sense think act that type of model is just referring to that little boomerang data come in cognitive processing action selection so we're in that figure the question was in order to measure surprise wouldn't we need another value of y i.e a separate y that includes prior belief great let's just assume that there is a generative model so the entity has like two parameters in its cognitive model the mean and the variance then y comes in and given the parameterization of the all that's needed is the data point to come in for the surprise to be calculated another value is needed whether we call it abc xyz z is just a mathematical abstraction so yes prior beliefs are important the prior beliefs that can be interpreted as the parameterizations of the cognitive model are required external signals is there a third y that is considered one could imagine a lot of like real world scenarios where a more complex model would be required you know two people looking at the thermometer or all these different sorts of situations but um unless somebody can unpack this a little more or explain what they were asking about then i don't think a third why is required mike yeah that was my question so i can try and unpack it um and so sticking with the thermometer example at the heart of that last question is if we have a thermometer that's registering the temperature can we consider that as sort of the the true observation that is the actual why um as measured by this instrument and then therefore we can compare our sort of internal why with the actual y so i have a really similar and related that's like two questions down um but it's in a different section but let me just like place some things on here and it goes back to like what we were saying about like what exactly is the hidden state so is the hidden state the temperature and the data is the reading from the thermometer that like that's like the the data y right like the out this this y in the middle but then like what is how i feel hot

or cold like i mean i'm perfectly positioned to be at 75 all the time so i know if it's like one degree too cold or one degree too hot like i've got to turn up the heat or turn it down or or whatever but like my registration my sensory input is not at 75 degrees like my sensory input is i'm hotter i'm cold and so this this is like where i mean we were talking about this yesterday in the math group like there's this really fuzzy line for me like i would love for someone to clarify that so i do get what you mean about like this in like shouldn't there be another y like there's the y out there in the world 75 degrees and then there's how i feel about that y okay thanks um variables are not like innately tagged with being observables or hidden states it's a model specific framing of what is going to be modeled as generated data and what is going to be modeled as a bayesian prior in multi-level bayesian modeling the priors themselves are generated from a higher or deeper level of the so one can be serving in fact multiple roles this is the minimal generated data expectation maximization type single layer bayesian kernel so we're going to think about this temperature example there's a hidden state of the generative process which is going to be the temperature of the world latent unmodeled so this isn't even claiming that there is such a thing as temperature this is just the latent unobserved temperature that is giving rise to thermometer readings which might have like different sorts of noise then there's another hidden state which is the evaluation of temperature this is just a schematic that's and also the whether it one is labeled x or y or l or triangle is like a norm and a convenience the letter doesn't matter itself it it will be used mostly consistently within the textbook but there's only so many letters and there's not a lot of coherence on notation use what else could be explored here yeah um mike and then ali so as i keep ruminating on this which is probably more than i should um it seems like there can be infinite y's right so um to the extent that you are taking action which will change your perception then you are as a result triggering potentially new whys in the system right so you just have this infinite potential for what y can be this speaks to the composability and the flexibility of active inference so like y could be one pixel of visual input it could be a 4k video why could be smell and a 4k video why could be lidar and this and that so why is just the generalized input of sense and then action might be related to this in a very direct way like y could be visual input and action could be your eye movement so that case is going to be explored a lot in the book however it could also be you're getting something in and then the actions are just totally unrelated maybe they don't affect the hidden state the causal process in the world at all or maybe actions enable different wise to enter the picture this is just the total essence kernel even trivial cases require some more apparatus yeah i have a question uh is it correct to say that the minimization of

variational free energy aka is exactly equivalent to the maximization of expectation uh i mean is it uh i mean the holy linear relationship between these two uh or possibly we have some uh kind of plateaued areas between these two extremes so this is something that we'll come to next week in our discussions on chapter two so we have not even discussed variational free energy yet today we talked about starting with a low road on some of the atomic calculations that are going to come into play like surprise and bayesian surprise in a bayesian framework and beginning to partition active entities and their action perception loops in terms of a bayesian graph that's going to be amenable to flexible modeling of perceptive cognitive active and out there in the world variables variables with those interpretations and then variational free energy is going to come into play as a way to bound so let's have some questions in discourse and we'll come to it next week but it's an awesome question jessica hi yes i have a question um i guess related to this figure 2.0 let's say like um in regards to like how we interpret on like what we observe so like i understand like you have a prior that might you know you're anticipating you know something like some other people's behavior or like how things should be that might be different from what you actually observe but a lot of times at least in terms of like when we're thinking like human beings like it's like an action in the world like how you interpret it and has to do a lot with on your own like views or like your own experiences and maybe that connects to the priors and like like the interpretation might be very different like from people to people the interpretation will vary a lot because everybody has on like a lot of like different experiences so those could be like uh encoded i guess in the priors um but it's not connected per say to the production part or maybe it is so i'm trying to kind of connect that idea and of how like our personal interpretation based on our experiences which could be priors make us see the like what's happening in the world differently than other people and so that why would look very different from you than for me just because we have lived different experiences so it's not like a like an actual why like it's not like you cannot really say this is a fact and because it's an interpretation at the end of the day great question and points yeah understanding how the individual setup of a given entity is related to its past experiences and how that could be modeled by priors is an important area and um some of those rich dynamics are not in this minimal nucleus but it'll be awesome to start to think about what does have to be in the box here to give rise to those kinds of dynamics um so that's going to end this discussion next week we will also be staying in and taking it more towards variational free energy and expected free energy so that should be fun we're going to in this room transfer immediately to the dot tools meeting so if you want to join for activelab.tools

meeting everyone is welcome and there's no like prerequisites or anything like that if you want to hang around and join tools stay in this exact gather space if you want to continue talking with other people about the book or anything else just head up into one of the rooms above and if anyone anyone who wants to can just continue any discussion that they want but in this space we're going to continue now with tools thanks everyone